1. Problem and audience

Problem

- Journaling is easy to start, but most people stop, and almost nobody rereads old entries.
- So the patterns stay hidden: what actually affects my mood, sleep and stress?
- Journal apps mostly store entries. Chatbots reply to one message but don’t remember me or follow up.

Opportunity

- An LLM with long-term memory can read every entry, find patterns across weeks, and check in on goals.

For whom

- Anyone who wants to keep a journal but hasn’t been able to stick with it.
- I’m the first user: a student under deadline pressure.

2. Proposed solution

WHAT I WILL BUILD

Reflect: a web app where you write a short daily journal entry, and an AI agent turns your entries into a weekly review.

Major capabilities

- Reads each entry and extracts mood, emotions, sleep, activities and people
- Keeps a long-term memory of all entries and goals
- Finds patterns across weeks, e.g. low mood after late nights
- Writes a weekly review with dates as evidence, plus one suggestion
- Follows up on that suggestion in later entries

What a user can do

1. Write a 2-minute entry each night
2. Answer one follow-up question when something is missing
3. Read the Sunday review and see mood and sleep trends

3. Approach

Technologies	Python, a SQLite database for long-term memory, and a Streamlit web interface.
Models	OpenAI gpt-4o-mini with JSON output, for reading entries, asking follow-up questions and writing reviews. Pattern statistics are computed in plain Python, not by the model.
Libraries	openai, streamlit, pandas, sqlite3
Data sources	• My own journal, 3+ weeks • A 21-entry test journal I wrote and labeled by hand • GoEmotions (Google, 2020): 58k Reddit comments labeled with 27 emotions + neutral
External systems	OpenAI API only. Entries are stored locally; only the entry text is sent to the API.

4. Scope and risks

What I can realistically complete

- ✓ Daily entry → extraction → memory
- ✓ Weekly review with patterns and one suggestion
- ✓ Follow-up on that suggestion in later entries
- ✓ A simple web app
- ✓ Evaluation on the test journal and GoEmotions

- ☆ If time allows: voice entries

Uncertainties

⚠ **Extraction accuracy**
The model may misread sleep times or moods.

⚠ **Too little data**
3 weeks may not be enough to find reliable patterns.

⚠ **Consistency**
User may not journal every day.

⚠ **Privacy and sensitive content**
Entries are personal and sometimes emotional.

5. Plan and success criteria

Major milestones

Proposal (Oct 29)

Prototype pipeline runs end to end on the test journal

Midpoint

Run the real model, measure accuracy, fix errors, start journaling daily

Final

3+ weeks of my own entries, finished app and README

How I will show it works

- Extraction accuracy against my hand labels on the 21 test entries
- Emotion accuracy on GoEmotions, compared with the 33% “always neutral” baseline
- Live demo: write an entry → follow-up question → weekly review on real data
- 3 weeks of my own use: do the patterns match what I actually experienced?